



KNS INSTITUTE OF TECHNOLOGY

DEPARTMENT OF PHYSICS

QUESTION BANK

22 SCHEME – FOR CSE/CSDS/ISE/AIIML STREAM

Sem/ Sec	1 Year CSE	SUBJECT CODE	BPHYS102 / BPHYS202	SUBJECT NAME	APPLIED PHYSICS FOR CSE STREAM
Faculty Name	M. Munavar Hussain T.N. Silpa K. Rafeeqe Ahmed				

COURSE OUTCOMES:

- C01: Describe the principles of LASERS and Optical fibers and their relevant applications.
- C02: Discuss the basic principles of the Quantum Mechanics and its application in Quantum Computing.
- C03: Summarize the essential properties of superconductors and its applications in qubits.
- C04: Illustrate the application of physics in design and data analysis.
- C05: Practice working in groups to conduct experiments in physics and perform precise and honest measurements.
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MODULE WISE IMPORTANT QUESTIONS

NOTE: The QUESTIONS which is Bold, needs more important.

MODULE- 1 -LASERS AND OPTICAL FIBRES

1. Define the terms Population inversion and Meta stable state. Derive the expression for energy density of radiation using Einstein's coefficients. Compare the expression with Planck's equation? (Jan 2019-18S) (July 2019-18S) (Jan 2020-18S) (Aug 2020-18S) (Jan 2021-18S) (July 2021-18S) (Feb 2022-18S) (July 2022-18S) (Feb 2022-21S) (July 2022-21S)
2. Describe the construction and working of Semiconductor laser with the help of energy band diagram. (Jan 2020-18S) (Jan 2021-18S) (July 2022-18S)
3. Explain the process of spontaneous and stimulated emission. OR Explain about the interaction of radiation with matter. (Jan 2018-17S) (Jan 2020-18S) (July 2021-18S)
4. Explain the requisites and conditions of a laser system. (July 2022-21S)
5. Define Acceptance Angle, Numerical aperture and Fractional index change Δ ? Obtain an expression for numerical aperture of an optical fibre with suitable diagram. (Jan 2018-17S) (Jan 2019-18S) (July 2019-18S) (Jan 2020-18S) (Aug 2020-18S) (July 2021-18S) (Feb 2022-18S) (July 2022-18S) (July 2022-21S)

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6. Explain the different types of optical fibres with suitable diagrams. (OR) Distinguish between the types of optical fibres based on Refractive Index profile and number of modes of propagation. (Jan 2018-17S) (Jan 2019-18S) (July 2019-18S) (Jan 2020-18S) (Aug 2020-18S) (July 2021-18S) (July 2022-18S)
7. Describe point-to-point communication system of an optical fibre with the help of a block diagram. (Jan 2021-18S) (Feb 2022-21S)
8. What is meant by Attenuation and attenuation coefficient? Explain the different attenuation mechanism. (OR) Mention the expression for attenuation co-efficient in a fibre of length L. Mention any two merits and demerits of an optical fibre. (Jan 2021-18S) (Feb 2022-18S)

MODULE- 2 - QUANTUM MECHANICS

1. State and explain the Heisenberg uncertainty principle. Using this principle, show that the electrons cannot reside in an atomic nucleus. (Jan 2018-17S) (Jan 2019-18S) (July 2019-18S) (Jan 2020-18S) (Aug 2020-18S) (Jan 2021-18S) (July 2021-18S) (Feb 2022-18S) (Feb 2022-21S) (July 2022-21S)
2. Derive time-independent Schrödinger wave equation. (Jan 2018-17S) (Jan 2019-18S) (Jan 2021-18S) (July 2022-18S) (Feb 2022-21S) (July 2022-21S)
3. Write down the Schrödinger equation for a particle in one-dimensional box. Obtain the Eigen functions and Eigen values for this particle with suitable waveforms and hence obtain the normalized wave equation. (July 2019-18S) (Jan 2020-18S) (Aug 2020-18S) (July 2021-18S) (Feb 2022-18S) (Feb 2022-21S) (July 2022-21S)
4. State and explain de Broglie's hypothesis.
5. Write a short note on wave function and Principle of complementarity.

MODULE- 3 – QUANTUM COMPUTING

1. Explain the representation of qubit using Bloch Sphere.
2. Explain Moore's law & its end.
3. Describe the working of controlled-Z gate mentioning its matrix representation and truth-table.
4. Elucidate the differences between classical and quantum computing.
5. State the Pauli matrices and apply Pauli matrices on the states $|0\rangle$ and $|1\rangle$.
6. Discuss the CNOT gate and its operation on four different input states.
7. Explain the Matrix representation of 0 and 1 States and apply identity operator I to $|0\rangle$ and $|1\rangle$ states.
8. Define a bit and qbit and explain the properties of qubit.
9. Explain pauli X,Y and Z gate on the states 0 and 1 with proper gate representations and their truth table.
10. Discuss the working of phase gate mentioning its matrix representation and truth table.
11. Explain Orthogonality and Orthonormality with an example for each.
12. Explain Single qubit gate and multiple qubit gate with an example for each.

MODULE- 4 -ELECTRICAL PROPERTIES OF MATERIALS AND APPLICATIONS

1. Write the assumptions of quantum free electron theory. Discuss about the success (merits) of quantum free electron theory. (OR) Describe how quantum free electron theory has been successful in overcoming the failures of classical free electron theory. (Jan 2019-18S) (July 2019-18S) (Jan 2020-18S) (July 2021-18S) (Feb 2022-18S) (July 2022-18S) (Feb 2022-21S) (July 2022-21S)
2. Discuss the merits of quantum free electron theory. (Jan 2018-17S) (Aug 2020-18S)
3. Explain Fermi energy and Fermi factor. Discuss the variation of Fermi energy with temperature. (Jan 2018-17S) (Jan 2019-18S) (July 2019-18S) (Aug 2020-18S) (Jan 2021-18S) (July 2021-18S) (Feb 2022-18S) (Feb 2022-21S)
4. Explain classical free electron theory of metals. Discuss in details about the failures of classical free electron theory. (Aug 2020-18S)
5. Discuss about the concept of phonon and Matheissen rule.
6. Explain in detail about Type-I and Type-II superconductors.
7. Explain Meissner's Effect and the variation of critical field with temperature.
8. Explain the phenomenon of superconductivity and discuss qualitatively the BCS theory of superconductivity for negligible resistance of metal at temperatures close to absolute zero.
9. Give the qualitative explanation of RF Squid with the help of a neat sketch.
10. Explain DC and AC Josephson effects and mention the applications of superconductivity in quantum computing.
11. Describe Meissner's Effect and hence classify superconductors into Soft and Hard superconductors using M-H graphs.

MODULE- 5 – PHYSICS OF ANIMATION

1. Discuss timing in linear motion, Uniform motion, slow in and slow out.
2. Sketch and explain the motion graphs for linear, easy ease, easy ease in and easy ease out cases of animation.
3. Distinguish between descriptive and inferential statistics.
4. Illustrate the odd rule and odd rule multipliers with a suitable example.
5. Describe Jumping and parts of jump.
6. Discuss the salient features of Normal distribution using bell curves.
7. Elucidate the importance of size & scale and weight and strength in animations.
8. Mention the general pattern of Monte-Carlo method and hence determine the value of π .
9. Describe the calculation of Push time and stop time with examples.
10. Discuss Poisson distribution and modelling the probability for proton decay.
11. Define a frame and show the importance of Frames per second in animation.

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ALL THE BEST

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